Survey of the Performance of Classification Algorithms on Heart Disease Datasets: A Systematic Literature Review

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***Abstract*—Machine learning is widely applied in medical informatics, especially for predicting heart disease. Despite numerous studies, there is a lack of systematic reviews that comprehensively map the classification algorithms used, datasets employed, and performance metrics evaluated within this field. This research provides a systematic review of the literature (SLR) focused on the performance of classification algorithms in predicting heart disease using real-world datasets. Adopting the PRISMA framework, we conducted a structured search across Scopus, IEEE Xplore, ACM, Springer, and ScienceDirect from January 2022 to December 2024. Forty-two peer-reviewed studies met our inclusion criteria. We extracted data on algorithm types, datasets, evaluation metrics, and feature engineering strategies. The most common datasets include Cleveland, Statlog, and Framingham, with Support Vector Machines (SVM) and Random Forest (RF) as dominant classifiers. However, algorithm performance varies significantly depending on feature selection and preprocessing techniques. This review identifies current research gaps, such as limited use of multiclass labels, insufficient real-clinical datasets, and lack of comparative benchmarks, suggesting directions for future research.**

Keywords— classification, heart disease, machine learning, systematic review, SVM, ensemble learning, PRISMA

I. INTRODUCTION

A diverse array of heart disease datasets is utilized in machine learning research, accessible from a variety of repositories, including the UCI Machine Learning Repository [1], Kaggle [2], PhysioNet [3], OpenML [4], and the National Institutes of Health (NIH) [5], among others.

This review incorporates datasets such as Cleveland, Statlog (Heart), Framingham, Hungarian, Z-Alizadeh Sani, Long Beach VA, and South African, each offering distinct features and demographics.

This paper centers on the Cleveland heart disease dataset, sourced from the UCI Repository, which comprises a well-organized structure of 76 attributes and a single class label. Since 2022, this dataset has been referenced as secondary data across 28 peer-reviewed studies, underscoring its pivotal role in heart disease research and its extensive applicability for developing predictive models. 2022.

Machine learning represents an advanced computational technique capable of aiding in the classification of heart disease [6]. Through computational approaches, machines can learn from datasets, made possible by machine learning algorithms [7]. In machine learning, classification aims to accurately predict the correct label for given input data [8]. However, not all machine learning classification models excel in accurately discerning classes; a significant limitation lies in their inability to extrapolate beyond existing data. Nonetheless, the scope of application can be expanded relatively easily by integrating new data points [9]. This limitation in class recognition presents an ongoing challenge for researchers, spurring efforts to optimize machine learning classification methods for improved accuracy and adaptability.

Optimization efforts in machine learning, particularly in classification, encompass various sophisticated strategies, including data preprocessing [10], feature selection [11], [12], ensemble learning [13], and deep learning techniques [14]. Additionally, established classification algorithms such as K-Nearest Neighbors, Decision Tree, Random Forest, Support Vector Machine (SVM), and Naïve Bayes remain integral to the field [15]. With continuous advancements in machine learning and artificial intelligence, these algorithms offer tailored solutions for diverse, complex contexts [7].

Classification algorithm development involves partitioning data into two primary sets: a larger training set, essential for model construction, and a test set, necessary for evaluating model performance and accuracy through predictive output.

This systematic literature review (SLR) synthesizes the latest advancements in classification algorithm performance on heart disease datasets, presenting a comprehensive view of the state-of-the-art algorithms and setting the foundation for emerging research directions. By emphasizing strategies such as preprocessing, targeted feature selection, and ensemble modeling, this SLR underscores the pathways to enhance model accuracy and robustness. These optimization approaches aim to maximize the efficacy of classification algorithms on heart disease datasets within the machine

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learning paradigm, thus facilitating more precise and reliable predictive outcomes in clinical and research applications.

This paper underscores the critical research avenues that warrant further exploration to optimize the performance of classification algorithms, employing the strategies previously discussed, with the overarching goal of identifying the most effective classification techniques. Through this systematic literature review (SLR), we provide an in-depth overview of machine learning methodologies for heart disease prediction.

The paper is structured as follows. Section I introduces the theoretical background of machine learning and its application to heart disease algorithms. Section II, Background Theory, establishes the foundational scientific concepts underpinning the study. Section III, Methods, outlines the detailed methodological framework guiding the literature exploration. Section IV, Results and Discussion, presents the findings, offering a nuanced analysis of the data. Lastly, Section V, Conclusion, synthesizes key insights and delineates prospective research directions within this evolving field..

II. BACKGROUND TEORY

A. Machine Learning

Machine learning (ML) is a key branch of artificial intelligence (AI) [16]–[18] Which focuses on creating advanced algorithms that allow computers to independently learn and make decisions based on data [17]. The primary aim of ML is to discern patterns within datasets [19], which can subsequently be leveraged for prediction, classification, or deriving insights with minimal human intervention. Challenges across diverse domains that prove difficult to resolve through traditional computational methods often become more manageable through using machine learning methods [20]. The ML process necessitates a structured learning phase utilizing training data. Following this, the model generated from the algorithm undergoes rigorous testing and validation against a previously unencountered test dataset. The performance of this model is assessed based on its accuracy, which is quantified by comparing the predicted outcomes against the actual values [21], [22]. This systematic approach not only enhances predictive capabilities but also contributes to the ongoing refinement of ML methodologies in various application.

B. Machine Learning Classification

Classification is a core task in machine learning [16], [17]. It is a key component of supervised learning techniques, characterized by the availability of labeled data [23]. In this context, the model represents trained on input features associated with the relevant output labels [16]. The foundational concept in classification begins with the number of classes: a binary class scenario involves the model classifying data into one of two possible categories, while a multi-class scenario requires the model to assign data to one of several categories. Following this, the process involves ensuring the availability of training data, where the model learns from labeled examples, and subsequently testing the model on previously unseen data. The last step consists of assessing the model's performance using standard evaluation metrics [16], [17], [24]. This systematic approach is essential for evaluating the effectiveness and precision of classification models across different applications.

C. Evaluation model

To assess the performance of machine learning classification models, several tools are utilized, one of which is the confusion matrix [17], [25]. It presents a tabular layout that displays the intersection of rows and columns, referred to as cells; each cell indicates the predicted values. This method assesses the performance of the classification model by measuring how well it aligns the predicted results compared to the actual outcomes. By analyzing the various components with analyzing the confusion matrix—containing true positives, false positives, true negatives, and false negatives—researchers can compute valuable metrics like accuracy, precision, recall, and F1 score, offering a thorough assessment of the model's performance.

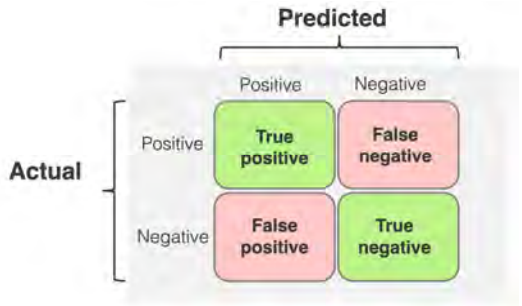


Fig. 1. Confusion matric

The following method is Accuracy [17], [26]–[28]

$$\text{Accuracy, } acc \frac{TP+TN}{TP+TN+FP+FN} , \quad (1)$$

$$\text{Precision, } prec \frac{TP}{TP+FP} \quad (2)$$

$$\text{Recall, } rec \frac{TP}{TP+FN} \quad (3)$$

$$\text{F1-Score } F1-score = 2 \frac{prec.recall}{Precision+recall} \quad (4)$$

In addition to the aforementioned metrics, graphical model evaluation techniques such as the Receiver Operating Characteristic (ROC) curve [29] are used to display performance. The ROC curve shows the connection between sensitivity (True Positive Rate, TPR) and 1-specificity (False Positive Rate, FPR). In other words, the ROC curve aids in understanding the balance between sensitivity and specificity for a particular model.

III. METHODS

A. Research Question (RQ)

At this stage, the research questions are identified as follows:

RQ1 : What heart disease datasets have been most commonly used in classification research between 2022 and 2024?

RQ2 : What feature sets are frequently selected from these datasets, and how do they impact model performance?

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RQ3 : Which classification algorithms demonstrate superior accuracy in recent studies on heart disease prediction?

RQ4 : What are the prevailing trends, challenges, and research gaps in applying machine learning to heart disease classification?

B. Review Protocol and Search Strategy

This SLR follows the PRISMA 2020 guidelines [30]. Literature was retrieved from five major databases: Scopus, IEEE Xplore, ScienceDirect, SpringerLink, and ACM Digital Library. The search focused on articles published between January 2022 and December 2024. The following search string was used:
("heart disease" OR "cardiac disease") AND ("classification algorithm" OR "machine learning") AND ("accuracy" OR "performance")

C. Criteria for Inclusion and Exclusion

To maintain the quality and relevance of the studies incorporated in this review, criteria for inclusion and exclusion were applied. Studies were included if they were peer-reviewed journal or conference publications, Concentrated on classifying heart disease through machine learning algorithms, presenting performance metrics such as accuracy, precision, and F1-score.

Conversely, studies were excluded if they were not written in English, categorized as editorial notes or narrative reviews, or lacked sufficient methodological detail such as the description of the dataset, algorithm implementation, or evaluation process.

D. Study Selection and Screening Process

A total of 241 articles were initially retrieved. After removing duplicates (n=32), 209 papers remained for abstract and title screening. 83 articles were excluded during this screening due to irrelevance to the research focus, such as not addressing classification algorithms or heart disease.

Of the remaining 126 full-text articles assessed for eligibility, 84 were excluded for the following reasons, 32 articles lacked sufficient methodological detail, 27 did not report quantitative performance metrics, 15 were general reviews not focused on classification algorithms, 10 were non-English or had inaccessible full texts.

Finally, 42 articles satisfied the inclusion criteria and were incorporated into the final synthesis. The Fig. 2 illustrates the detailed process of study identification, screening, exclusion, and inclusion based on the PRISMA 2020 guidelines [30]. It includes reasons and counts for exclusions at each stage to improve transparency and reproducibility.

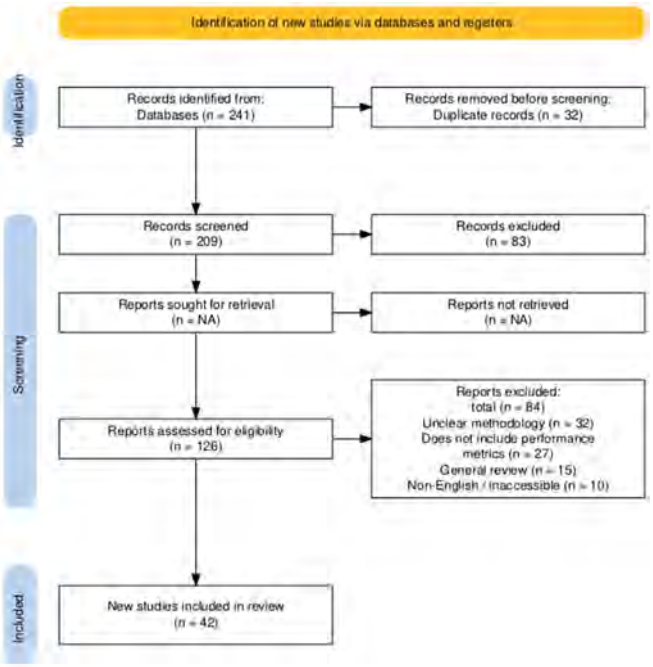


Fig. 2 PRISMA Flow Diagram

E. Quality Assessment

To evaluate the methodological rigor of the included studies, a structured quality assessment was conducted based on four criteria: (1) use of a valid and publicly recognized dataset; (2) clear definition and explanation of algorithm and methodological procedures; (3) provision of performance evaluation metrics such as accuracy or F1-score; and (4) sufficient documentation to ensure reproducibility of results.

Each study was scored on a binary scale (0 or 1) for each criterion, resulting in a total quality score ranging from 0 to 4. Only studies with a minimum score of 3 were considered eligible for inclusion in this review to ensure robustness and reliability of the synthesized findings

F. Data Extraction

A standardized approach to data extraction was implemented to ensure uniformity and comprehensiveness. For each selected study, the following elements were systematically recorded: the names of the authors, year of publication, dataset(s) used, feature selection techniques applied, classification algorithm(s) employed, presented performance metrics such as accuracy, precision, recall, and F1-score, and any noted limitations or unique contributions. These extracted attributes formed the basis for the synthesis of results and the comparative analysis presented in Section IV

IV. RESULT AND DISCUSION

A. Dataset Utilization (RQ1)

Among the 42 studies, the Cleveland dataset was the most commonly utilized dataset (n=26), followed by Statlog (n=10), Framingham (n=6), and Z-Alizadeh Sani (n=4). Many studies favored Cleveland due to its availability and

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manageable number of attributes. The Cleveland dataset, created by the Robert Detrano team at the Cleveland Clinic Foundation, is accessible via the UCI Machine Learning Repository, has become the de facto benchmark dataset for heart disease prediction due to several factors. First, it offers a well-structured and relatively clean set of attributes, with 76 features initially collected, 14 of which are most commonly used in ML studies [6], [12]. Second, its use in many prior peer-reviewed studies enables comparative evaluation and benchmarking across research works [31], [32].

Furthermore, its moderate sample size (303 instances) makes it computationally efficient for training a wide range of ML models, from logistic regression to deep learning, without necessitating excessive computing resources. Several works [33], [34] also highlight the dataset's compatibility with feature selection techniques and its adaptability to both binary and multiclass classification scenarios. Due to these reasons, the Cleveland dataset remains the preferred choice in recent literature.

TABLE 1 Datasets used in Research

Dataset	General information sources, amount of data, features	Total	Paper
Cleveland	UCI, 303, 14	28	[35]–[62]
Statlog (Heart)	UCI, 270, 13	8	[59], [60], [63], [64], [61], [47], [62], [65]
Framingham	UCI, 5000, 15	5	[55], [66], [67], [47], [58]
Hungarian	UCI, 294, 13	5	[35][68][56][36][57]
Z-Alizadeh Sani	UCI, 303, 56	3	[47], [69], [70]
Long Beach VA	Long Beach Veterans Affairs Medical Center, 200, 15	2	[36], [56]
South African	UCI, 462, 9	1	[47]

Table 2 summarizes the data extraction and quality assessment process for the 42 studies included in this review. It includes information on dataset(s) used, classification algorithm(s) employed, feature selection methods, reported performance metrics, and the quality score for each study based on a 4-point checklist. Notably, 10 of the 42 studies achieved an average quality score above 3, highlighting their methodological rigor and reliability. This allows for a comparative analysis of the studies' methodologies and performance.

TABLE 2 Data Extraction and Quality Assessment Table

No	Author (Year)	Dataset	Algorithm	Accuracy	Feature Selection	Quality Score (0-4)
1	Ram Kumar (2024)	Cleveland	KNN, SVM, ANN, CNN	83.6%	Not stated	3
2	Komal Kumar (2024)	Cleveland	LR, SVM, NB, KNN	89.9%	RFE	4
3	Ozcan & Peker (2023)	Cleveland, Hungary, Statlog	CART	87%	Not stated	3
4	Khan et al. (2023)	Khyber Hospital	DT, RF, NB, SVM	75-85%	Not stated	3
5	Dalal et al. (2023)	UCI, Kaggle	QUEST, RF, NN	99.1%	PCA	4
6	Chandrasekhar (2023)	Cleveland	RF, KNN, AdaBoost	93.4%-95%	ReliefF	4
7	Ullah et al. (2022)	Cleveland	DT, SVM, LR	97.9%	Manual	4
8	Saboor et al. (2022)	Multiple	RF, GA, NB, SVM	83.6%-96.7%	GA	4
9	Bora et al. (2022)	Cleveland	RF, GBoost, NB	73%-96%	LASSO	4
10	Ambrish G (2022)	Cleveland, Framingham	LR, RF, SGB	87.1%	Embedded	4

B. Feature Selection and Dataset Attributes (RQ2)

Although the Cleveland dataset contains 76 features, most studies use only 13–14 key attributes. Typical features consist of age, sex, chest pain type, resting blood pressure, serum cholesterol, and maximum heart rate. Techniques for feature selection, including Recursive Feature Elimination (RFE), Genetic Algorithms (GA), and Principal Component Analysis (PCA), are employed to reduce dimensionality and improve model performance. Studies that applied feature selection methods generally reported increased accuracy, often by 2–10%, compared to models that utilize the complete set of features. For example, models that implemented RFE on the Cleveland dataset achieved accuracy improvements from ~85% to 95%. This suggests that selecting relevant and non-redundant attributes significantly contributes to model generalizability and computational efficiency.

Notably, a study by Komal Kumar et al. [35] implemented RFE in combination with SVM and reported an accuracy of 89.91%, outperforming configurations without feature selection. Similarly, Chandrasekhar and Peddakrishna [34] combined Relief F-based selection with ensemble classifiers and achieved up to 95% accuracy. These examples illustrate the importance of carefully engineered feature subsets in maximizing model performance while reducing computational costs.

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TABLE 3 14 key attributes

No	Variable Name	Role	Type	Description
1	age	Feature	Integer	
2	sex	Feature	Categorical	
3	cp	Feature	Categorical	
4	trestbps	Feature	Integer	Resting blood pressure (at the time of hospital admission)
5	chol	Feature	Integer	Serum cholesterol levels
6	fbs	Feature	Categorical	Fasting blood sugar levels exceeding 120 mg/dl
7	restecg	Feature	Categorical	
8	thalach	Feature	Integer	Peak heart rate achieved
9	exang	Feature	Categorical	Angina triggered by exercise
10	oldpeak	Feature	Integer	Exercise-induced ST depression compared to rest
11	slope	Feature	Categorical	
12	ca	Feature	Integer	Number of major vessels (0-3) highlighted by fluoroscopy
13	thal	Feature	Categorical	
14	num	Target	Integer	Heart disease diagnosis

C. *Algorithm Performance (RQ3)*

SVM (n=15) and RF (n=13) are the most frequently used classifiers, followed by Logistic Regression, KNN, and Naïve Bayes. Ensemble techniques like Gradient Boosting, XGBoost, and AdaBoost demonstrated improved accuracy when paired with feature selection. However, CNNs and Deep Neural Networks are underexplored due to dataset size limitations.

In terms of performance, SVM classifiers achieved accuracies ranging between 85% and 95% across various studies. For instance, Ullah et al. [71] reported an SVM accuracy of 97.9% using the Cleveland dataset. Random Forest classifiers also demonstrated strong performance, with accuracies typically falling between 88% and 95%, such as in the work of Chandrasekhar and Peddakrishna [34]. Logistic Regression models showed slightly lower accuracy, generally ranging from 80% to 89%, while KNN and Naïve Bayes varied between 75% and 90% depending on the dataset and preprocessing applied.

Boosting algorithms reported some of the highest accuracies when combined with optimized feature selection. Dalal et al. [72] achieved 99.1% accuracy using a combination of QUEST, Random Forest, and Neural Networks. Similarly, the use of XGBoost and Gradient Boosting in studies by Bora et al. [73] and Budholiya et al. [44] resulted in accuracies between 95% and 98%.

Fig. 3 illustrates the reported accuracy ranges for each classification algorithm, highlighting how ensemble methods generally outperform traditional models when appropriately tuned and combined with effective feature engineering strategies

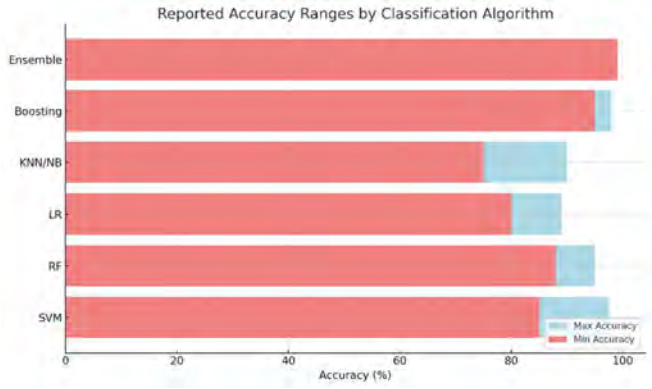


Fig. 3 Reported Accuracy Ranges by Classification Algorithm (*SVM: 85–97.9%, RF: 88–95%, LR: 80–89%, KNN/NB: 75–90%, Boosting: 95–98%, Ensemble: up to 99.1%*)

These results suggest that while traditional algorithms like SVM and RF remain highly reliable, ensemble techniques—especially when paired with effective feature engineering—offer significant advantages in predictive accuracy. While traditional algorithms like SVM and RF remain highly reliable, ensemble techniques—especially when paired with effective feature engineering—offer significant advantages in predictive accuracy.

D. *Trends, Gaps, and Challenges (RQ4)*

The examination of the chosen studies highlights various trends, challenges, and research gaps in the field of heart disease classification through machine learning. One prominent trend is the dominant reliance on open-access datasets such as Cleveland and Statlog. These datasets are widely used due to their accessibility and structured formats, enabling comparative evaluations across different studies. In terms of classification tasks, most research is focused on binary classification (i.e., forecasting the presence or absence of heart disease), with limited exploration of multiclass classification that could reflect various stages or types of heart conditions. Furthermore, the emergence of ensemble learning and hybrid modeling techniques has gained traction, particularly in achieving improved accuracy and robustness.

Despite these advancements, several gaps persist. Many studies still depend on benchmark datasets rather than real clinical data sourced directly from hospitals or health institutions, limiting the generalizability of their findings. Moreover, the interpretability of machine learning models is still insufficiently explored, with limited studies utilizing explainable AI (XAI) methods to enhance the transparency and comprehensibility of predictions for medical professionals.

Major obstacles also impede further advancement, such as the problem of imbalanced class distributions in datasets, which can bias classifiers toward the majority class, and the absence of standardized benchmarks that complicate performance comparisons. Another challenge is the inconsistency in data preprocessing and feature engineering practices, which affects reproducibility across studies. Addressing these limitations through more diverse datasets,

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standardized methodologies, and the integration of interpretability tools is essential for advancing reliable and clinically applicable ML-based heart disease prediction systems

E. Related research

Table 3 presents a comprehensive list of studies conducted between 2022 and 2024 that focus regarding the performance of classification algorithms used on heart disease datasets.

TABLE 4 Research on the performance of classification algorithms applied to heart disease datasets.

Authors and Year of publish	Dataset	Methods	Result
R Ram Kumar at al 2024 [54]	Cleveland	KNN, SVM, ANN, CNN	83.61% 66.7% 74.2% 70.08%
Komal Kumar at al 2024[35]	Cleveland	LR, SVM, NB, KNN, SGD, DT, ADB, MLP	89,91%
Ozcan and Peker 2023[32]	Cleveland, Hungarian, Switzerland, Long Beach VA, and Statlog	Classification and Regression Tree (CART)	87%
Khan et al 2023 [74]	Khyber Teaching Hospital and Lady Reading Hospital in Pakistan	DT, RF, LR, NB, SVM	83% 85% 83% 75% 83%
Dalal et al 2023[72]	Compare UCI and Kaggle Dataset	QUEST, RF, NN, BN	99.1%
Chandrasekhar and Peddakrishna 2023[34]	Cleveland	RF, KNN, LR, NB, Gradient Boosting, AdaBoost	93.44% 95.00%
Ullah et al 2022[71]	Cleveland	DT, NN, SVM, KNN, LR.	97.9%
Saboor et al 2022[33]	UCI, Z-Alizadeh Sanitation Heart, Hungarian and Long Beach VA, Cleveland, Hungary, Switzerland	KNN, NN, SVM, GA, J48 (DT), RF, NB	83.66% to 96.72%
Bora et al 2022 [73]	Cleveland	LR, SVM, RF, NB, KNN, GBoost, XGBoost	73% to 96%
Ambrish G et al 2022[75]	Cleveland, Framingham	LR, SVM, RF, SGB, J48, MP	87.10%
AHMAD et al 2022[76]	Heart Statlog, Cleveland Hungary, Heart Disease Clinical Record Data Set 2020	LDA, RF, GBC, DT, SVM, KNN	100% 84.09% 100% 95.78% 85.71%

V. CONCLUSION

This systematic review offers a thorough synthesis of recent studies on the effectiveness of classification algorithms used on heart disease datasets. The Cleveland dataset became the most commonly used source, primarily due to its structured features, moderate size, and established benchmarking potential. Feature selection techniques such as RFE, PCA, and GA were shown to enhance classification accuracy by up to 10%, with studies demonstrating substantial improvements when these methods were used.

Support Vector Machine (SVM) and Random Forest (RF) remain the most frequently implemented algorithms, offering robust performance with accuracies ranging between 85–97.9% and 88–95%, respectively. Ensemble and boosting methods such as XGBoost and hybrid approaches showed superior accuracy, with some reaching 99.1% when combined with optimized feature engineering.

Emerging trends include an increased focus on ensemble learning, while challenges persist around class imbalance, lack of real clinical datasets, and the limited use of explainable AI techniques. This review highlights critical research gaps such as the need for multiclass classification studies, model transparency, and standardized benchmarking frameworks.

Future work should prioritize expanding dataset diversity, utilizing models that are interpretable and explainable, assessing models in multiclass scenarios or real-time clinical environments. Such advancements are essential to enhanced the reliability, generalizability, and the clinical applicability of machine learning-based heart disease prediction systems.

REFERENCES

[1] UCI, “UCI Machine Learning Repository,” 2023. [Online]. Available: https://archive.ics.uci.edu/ml/index.php.

[2] Kaggle, “Heart Disease UCI,” 2023. [Online]. Available: https://www.kaggle.com/uciml/heart-disease-uci.

[3] PhysioNet, “Heart Disease Data,” 2023. [Online]. Available: https://physionet.org/static/published-projects/.

[4] OpenML, “Heart Disease Dataset,” 2023. [Online]. Available: https://www.openml.org/d/1464.

[5] National Institutes of Health, “Heart Disease Dataset,” 2023. [Online]. Available: https://www.nih.gov/.

[6] J. P. Li, A. U. Haq, S. U. Din, J. Khan, A. Khan, and A. Saboor, “Heart disease identification method using machine learning classification in e-healthcare,” *IEEE access*, vol. 8, pp. 107562–107582, 2020.

[7] I. Ibrahim and A. Abdulazeez, “The role of machine learning algorithms for diagnosing diseases,” *J. Appl. Sci. Technol. Trends*, vol. 2, no. 01, pp. 10–19, 2021.

[8] P. Saraswat, “Supervised machine learning algorithm: A review of classification techniques,” *Integr. Emerg. Methods Artif. Intell. \& Cloud Comput.*, pp. 477–482, 2021.

[9] M. R. Dobbelaere, P. P. Plehiers, R. Van de Vijver, C. V Stevens,

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and K. M. Van Geem, "Machine Learning in Chemical Engineering: Strengths, Weaknesses, Opportunities, and Threats," *Engineering*, vol. 7, no. 9, pp. 1201–1211, 2021.
[10] R. Rajendran and A. Karthi, "Heart disease prediction using entropy based feature engineering and ensembling of machine learning classifiers," *Expert Syst. Appl.*, vol. 207, p. 117882, 2022.
[11] M. S. Pathan, A. Nag, M. M. Pathan, and S. Dev, "Analyzing the impact of feature selection on the accuracy of heart disease prediction," *Healthc. Anal.*, vol. 2, p. 100060, 2022.
[12] P. Ghosh *et al.*, "Efficient prediction of cardiovascular disease using machine learning algorithms with relief and LASSO feature selection techniques," *IEEE Access*, vol. 9, pp. 19304–19326, 2021.
[13] M. Hosni *et al.*, "A systematic mapping study for ensemble classification methods in cardiovascular disease," *Artif. Intell. Rev.*, vol. 54, pp. 2827–2861, 2021.
[14] A. Rath, D. Mishra, G. Panda, and S. C. Satapathy, "Heart disease detection using deep learning methods from imbalanced ECG samples," *Biomed. Signal Process. Control*, vol. 68, p. 102820, 2021.
[15] M. Bansal, A. Goyal, and A. Choudhary, "A comparative analysis of K-nearest neighbor, genetic, support vector machine, decision tree, and long short term memory algorithms in machine learning," *Decis. Anal. J.*, vol. 3, p. 100071, 2022.
[16] T. M. Mitchell and T. M. Mitchell, *Machine learning*, vol. 1, no. 9. McGraw-hill New York, 1997.
[17] C. M. Bishop and N. M. Nasrabadi, *Pattern recognition and machine learning*, vol. 4, no. 4. Springer, 2006.
[18] C. Krittanawong *et al.*, "Machine learning prediction in cardiovascular diseases: a meta-analysis," *Sci. Rep.*, vol. 10, no. 1, p. 16057, 2020.
[19] D. Sacha *et al.*, "What you see is what you can change: Human-centered machine learning by interactive visualization," *Neurocomputing*, vol. 268, pp. 164–175, 2017.
[20] D. Maulud and A. M. Abdulazeez, "A review on linear regression comprehensive in machine learning," *J. Appl. Sci. Technol. Trends*, vol. 1, no. 2, pp. 140–147, 2020.
[21] V. Sharma, S. Yadav, and M. Gupta, "Heart disease prediction using machine learning techniques," in *2020 2nd international conference on advances in computing, communication control and networking (ICACCCN)*, 2020, pp. 177–181.
[22] C. M. Bhatt, P. Patel, T. Ghetia, and P. L. Mazzeo, "Effective heart disease prediction using machine learning techniques," *Algorithms*, vol. 16, no. 2, p. 88, 2023.
[23] P. Cunningham, M. Cord, and S. J. Delany, "Supervised Learning," in *Machine Learning Techniques for Multimedia: Case Studies on Organization and Retrieval*, M. Cord and P. Cunningham, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2008, pp. 21–49.
[24] T. Hastie, R. Tibshirani, J. H. Friedman, and J. H. Friedman, *The elements of statistical learning: data mining, inference, and prediction*, vol. 2. Springer, 2009.
[25] S. V Stehman, "Selecting and interpreting measures of thematic classification accuracy," *Remote Sens. Environ.*, vol. 62, no. 1, pp. 77–89, 1997.
[26] P. Cichosz, "Assessing the quality of classification models: Performance measures and evaluation procedures," *Cent. Eur. J. Eng.*, vol. 1, pp. 132–158, 2011.
[27] A. Ng, "Machine learning yearning: Technical strategy for AI engineers, in the era of deep learning," 2018.
[28] E. Alpaydin, *Introduction to machine learning*. MIT press, 2020.
[29] T. Fawcett, "An introduction to ROC analysis," *Pattern Recognit. Lett.*, vol. 27, no. 8, pp. 861–874, 2006.
[30] N. R. Haddaway, M. J. Page, C. C. Pritchard, and L. A. McGuinness, "PRISMA2020: An R package and Shiny app for producing PRISMA 2020-compliant flow diagrams, with interactivity for optimised digital transparency and Open Synthesis," *Campbell Syst. Rev.*, vol. 18, no. 2, p. e1230, Jun. 2022.
[31] A. G, B. Ganesh, A. Ganesh, C. Srinivas, Dhanraj, and K. Mensinkal, "Logistic regression technique for prediction of cardiovascular disease," *Glob. Transitions Proc.*, vol. 3, no. 1, pp. 127–130, 2022.
[32] M. Ozcan and S. Peker, "A classification and regression tree algorithm for heart disease modeling and prediction," *Healthc. Anal.*, vol. 3, p. 100130, 2023.
[33] A. Saboor, M. Usman, S. Ali, A. Samad, M. F. Abrar, and N. Ullah, "A Method for Improving Prediction of Human Heart Disease Using Machine Learning Algorithms," *Mob. Inf. Syst.*, vol. 2022, no. 1, p. 1410169, Jan. 2022.
[34] N. Chandrasekhar and S. Peddakrishna, "Enhancing heart disease prediction accuracy through machine learning techniques and optimization," *Processes*, vol. 11, no. 4, p. 1210, 2023.
[35] K. K. Napa, A. K. Kumar, S. Murugan, K. Mahammad, T. A. Assegie, and K. Okokpujie, "Early prediction of chronic heart disease with recursive feature elimination and supervised learning techniques," *IAES Int. J. Artif. Intell.*, vol. 13, no. 1, pp. 730 – 736, 2024.
[36] G. N. Ahmad *et al.*, "Mixed Machine Learning Approach for Efficient Prediction of Human Heart Disease by Identifying the Numerical and Categorical Features," *Appl. Sci.*, vol. 12, no. 15, 2022.
[37] R. A. Kanza, M. U. H. Al Rasyid, and S. Sukaridhoto, "Efficient Early Detection of Patient Diagnosis and Cardiovascular Disease using an IoT System with Machine Learning and Fuzzy Logic," *Int. J. Comput. Digit. Syst.*, vol. 16, no. 1, pp. 183 – 199, 2024.
[38] S. Gandhi, T. Poongodi, and K. Sampath Kumar, "Enhancing data security of cardiac patients in IoMT with Twin-Shield Encryption," *J. Auton. Intell.*, vol. 7, no. 2, 2024.
[39] S. Alsubai, A. Alqahtani, A. Binbusayyis, M. Sha, A. Gumaei, and S. Wang, "Heart Failure Detection Using Instance Quantum Circuit Approach and Traditional Predictive Analysis," *Mathematics*, vol. 11, no. 6, 2023.
[40] G. Abdulsalam, S. Meshoul, and H. Shaiba, "Explainable Heart Disease Prediction Using Ensemble-Quantum Machine Learning Approach," *Intell. Autom. Soft Comput.*, vol. 36, no. 1, pp. 761 – 779, 2023.
[41] M. D. Fathima, S. J. Samuel, and S. P. Raja, "HDDSS: An Enhanced Heart Disease Decision Support System Using RFE-ABGNB Algorithm," *Int. J. Interact. Multimed. Artif. Intell.*, vol. 8, no. 2, pp. 29 – 37, 2023.
[42] P. A. Sadiyamole and S. Manju Priya, "A Robust Cardiovascular Disease Predictor Based on Genetic Feature Selection and Ensemble Learning Classification," *Int. J. Electr. Comput. Eng. Syst.*, vol. 14, no. 7, pp. 799 – 809, 2023.
[43] K. Chandrashekar and A. T. Narayanreddy, "An Ensemble Feature Optimization for an Effective Heart Disease Prediction Model," *Int. J. Intell. Eng. Syst.*, vol. 16, no. 2, pp. 517 – 525, 2023.
[44] K. Budholiya, S. K. Shrivastava, and V. Sharma, "An optimized XGBoost based diagnostic system for effective prediction of heart disease," *J. King Saud Univ. - Comput. Inf. Sci.*, vol. 34, no. 7, pp. 4514 – 4523, 2022.
[45] S. Christal, M. Sundararajan, G. U. Maheswari, P. Kaur, and A. Kaushik, "Heart Diseases Diagnosis Using Chaotic Harris Hawk Optimization with E-CNN for IoMT Framework," *Inf. Technol. Control*, vol. 52, no. 2, pp. 500 – 514, 2023.
[46] K. V. V. Reddy, I. Elamvazuthi, A. A. Aziz, S. Paramasivam, H. N. Chua, and S. Pranavanand, "An Efficient Prediction System for Coronary Heart Disease Risk Using Selected Principal Components and Hyperparameter Optimization," *Appl. Sci.*, vol. 13, no. 1, 2023.
[47] N. P and S. Narayan, "Cardiac disease detection using cuckoo search enabled deep belief network," *Intell. Syst. with Appl.*, vol. 16, p. 200131, 2022.
[48] B. A. Majeed, A. Y. Hardan, B. Y. Hardan, and D. F. Munaf, "Accurate AI-Based Chatbot to Diagnose Heart Diseases Pre-Human Doctor Consultation," *Rev. d'Intelligence Artif.*, vol. 38, no. 1, pp. 213 – 220, 2024.
[49] S. Meda and R. Bhogapathi, "An Integrated Machine Learning Model for Heart Disease Classification and Categorization," *J. Comput. Sci.*, vol. 18, no. 4, pp. 257 – 265, 2022.
[50] S. Bashir, I. U. Khattak, A. Khan, F. H. Khan, A. Gani, and M. Shiraz, "A Novel Feature Selection Method for Classification of Medical Data Using Filters, Wrappers, and Embedded Approaches," *Complexity*, vol. 2022, 2022.
[51] S. Das, S. K. Pradhan, S. Mishra, S. Pradhan, and P. K. Pattnaik,

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"DIAGNOSIS OF CARDIAC PROBLEM USING ROUGH SET THEORY AND MACHINE LEARNING," *Indian J. Comput. Sci. Eng.*, vol. 13, no. 4, pp. 1112 – 1131, 2022.

[52] S. Karthikeyini, G. Vidhya, T. Vetriselvi, and K. Deepa, "Heart Disease Prognosis Using D-GRU with Logistic Chaos Honey Badger Optimization in IoMT Framework," *Inf. Technol. Control*, vol. 52, no. 2, pp. 367 – 380, 2023.

[53] A. K. Berdaly and Z. M. Abdiahmetova, "PREDICTING HEART DISEASE USING MACHINE LEARNING ALGORITHMS," *Kazn. Bull. Math. Mech. Comput. Sci. Ser.*, vol. 115, no. 3, pp. 101 – 111, 2022.

[54] R. P. Ram Kumar *et al.*, "Investigations on cardiovascular diseases and predicting using machine learning algorithms," *Cogent Eng.*, vol. 11, no. 1, p. 2386381, 2024.

[55] S. C. M. Sundararajan, G. P. Bharathi, U. Loganathan, and S. Vadivel, "Improved smart healthcare system of cloud-based IoT framework for the prediction of heart disease," *Inf. Technol. Control*, vol. 52, no. 2, pp. 529–540, 2023.

[56] S. Molla *et al.*, "A predictive analysis framework of heart disease using machine learning approaches," *Bull. Electr. Eng. Informatics*, vol. 11, no. 5, pp. 2705–2716, 2022.

[57] A. K. Dubey, A. K. Sinhal, and R. Sharma, "An improved auto categorical PSO with ML for heart disease prediction," *Eng. Technol. \& Appl. Sci. Res.*, vol. 12, no. 3, pp. 8567–8573, 2022.

[58] B. A. M. Metwally, N. E. Mekky, and I. M. Elhenawy, "Heart Disease Prediction Using Genetic Algorithm with Machine Learning Classifiers," *Int. J. Adv. Sci. Eng. Inf. Technol.*, vol. 12, no. 5, pp. 1887 – 1894, 2022.

[59] K. Dissanayake and M. G. M. Johar, "Two-level boosting classifiers ensemble based on feature selection for heart disease prediction," *Indones. J. Electr. Eng. Comput. Sci.*, vol. 32, no. 1, pp. 381–391, 2023.

[60] C.-L. Cocianu, C. R. Uscatu, K. Kofidis, S. Muraru, and A. G. V\uaduva, "Classical, evolutionary, and deep learning approaches of automated heart disease prediction: a case study," *Electronics*, vol. 12, no. 7, p. 1663, 2023.

[61] A. K. Dubey, U. Gupta, and S. Jain, "Medical data clustering and classification using TLBO and machine learning algorithms," *Comput. Mater. Contin.*, vol. 70, no. 3, pp. 4523 – 4543, 2022.

[62] R. Aggarwal and S. Kumar, "Classification Model for Meticulous Presaging of Heart Disease Detection through SDA and NCA using Machine learning:CMSDANCA," *Int. J. Recent Innov. Trends Comput. Commun.*, vol. 10, no. 1, pp. 217 – 224, 2022.

[63] T. Karadeniz, H. H. Mara\cs, G. Tokdemir, and H. Ergezer, "Two majority voting classifiers applied to heart disease prediction," *Appl. Sci.*, vol. 13, no. 6, p. 3767, 2023.

[64] K. V. V. Reddy, I. Elamvazuthi, A. A. Aziz, S. Paramasivam, H. N. Chua, and S. Pranavanand, "An efficient prediction system for coronary heart disease risk using selected principal components and hyperparameter optimization," *Appl. Sci.*, vol. 13, no. 1, p. 118, 2022.

[65] A. Dubey, U. Gupta, and S. Jain, "Medical data clustering and classification using TLBO and machine learning algorithms," *Comput. Mater. Contin.*, vol. 70, no. 3, pp. 4523–4543, 2021.

[66] P. Gupta and D. Seth, "Comparative analysis and feature importance of machine learning and deep learning for heart disease prediction," *Indones. J. Electr. Eng. Comput. Sci.*, vol. 29, no. 1, p. 451, 2022.

[67] L. Riyaz, M. A. Butt, and M. Zaman, "A novel ensemble deep learning model for coronary heart disease prediction," *Rev. d'Intelligence Artif.*, vol. 36, no. 6, pp. 825–832, 2022.

[68] A. Ramesh, C. K. Subbaraya, and R. K. G. Krishnegowda, "A remote health monitoring framework for heart disease and diabetes prediction using advanced artificial intelligence model.," *Indones. J. Electr. Eng. Comput. Sci.*, vol. 30, no. 2, p. 846, 2023.

[69] J. J. Gabriel and L. J. Anbarasi, "Optimizing Coronary Artery Disease Diagnosis: A Heuristic Approach Using Robust Data Preprocessing and Automated Hyperparameter Tuning of eXtreme Gradient Boosting," *IEEE Access*, vol. 11, pp. 112988–113007, 2023.

[70] P. Anuradha and V. K. David, "Feature selection by ModifiedBoostARoota and classification by CatBoost model on high dimensional heart disease datasets," *Int. J. Comput. Theory Eng.*, vol. 14, no. 01, p. 2022, 2022.

[71] A. Ullah, S. A. Khan, T. Alam, S. Luma-Osmani, and M. Sadie, "Heart disease classification using various heuristic algorithms," *Int. J. Adv. Appl. Sci.*, vol. 2252, no. 8814, p. 8814, 2022.

[72] S. Dalal *et al.*, "Application of machine learning for cardiovascular disease risk prediction," *Comput. Intell. Neurosci.*, vol. 2023, no. 1, p. 9418666, 2023.

[73] N. Bora, S. Gutta, and A. Hadaegh, "Using machine learning to predict heart disease," California State University San Marcos, 2021.

[74] A. Khan, M. Qureshi, M. Daniyal, and K. Tawiah, "A novel study on machine learning algorithm-based cardiovascular disease prediction," *Heal. \& Soc. Care Community*, vol. 2023, no. 1, p. 1406060, 2023.

[75] G. Ambrish, B. Ganesh, A. Ganesh, C. Srinivas, K. Mensinkal, and others, "Logistic regression technique for prediction of cardiovascular disease," *Glob. Transitions Proc.*, vol. 3, no. 1, pp. 127–130, 2022.

[76] G. N. Ahmad, S. Ullah, A. Algethami, H. Fatima, and S. M. H. Akhter, "Comparative study of optimum medical diagnosis of human heart disease using machine learning technique with and without sequential feature selection," *ieee access*, vol. 10, pp. 23808–23828, 2022.

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